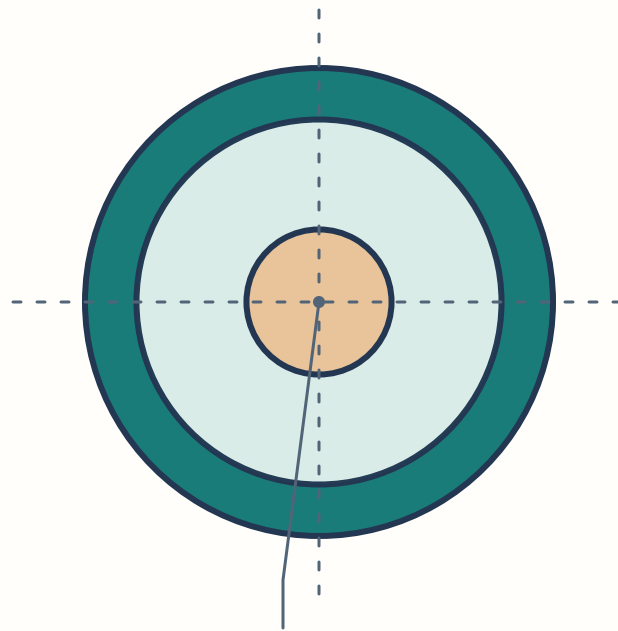


# How an optical fibre guides light

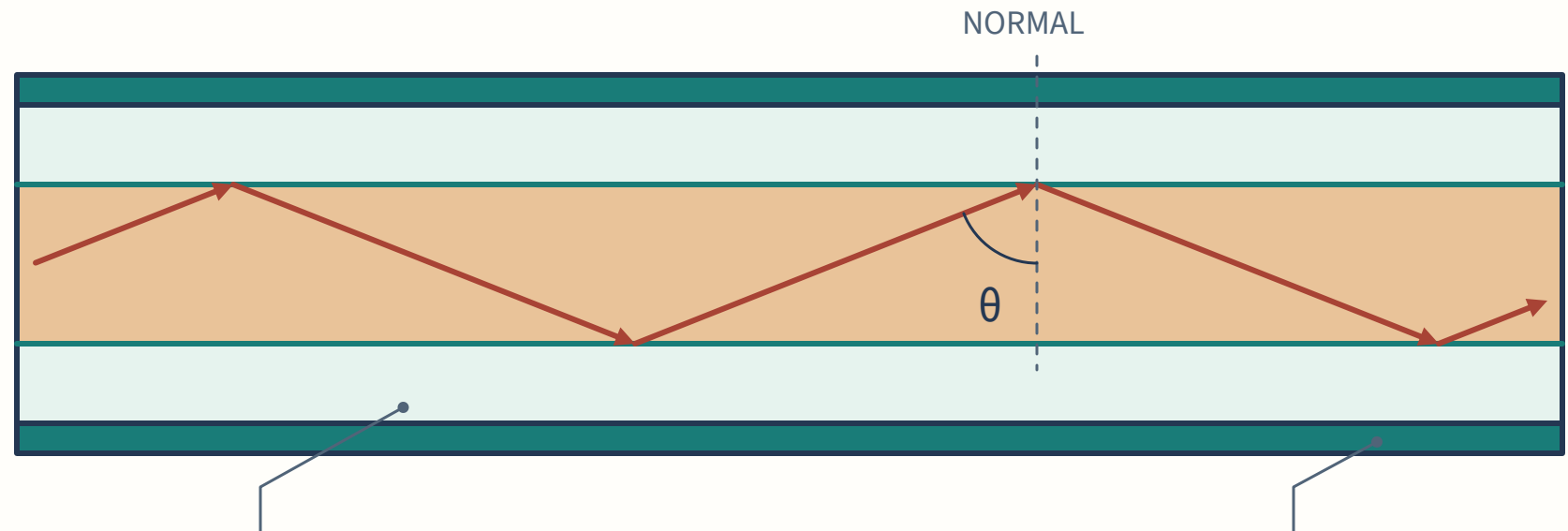
One explanation, three visual directions. A conceptual step-index fibre in the ray model.

A / END VIEW



Core ·  $n_1$

B / LONGITUDINAL SECTION



Cladding ·  $n_2$

Protective coating

## 01 / INDEX CONTRAST

$$n_1 > n_2$$

Total internal reflection occurs at the core-cladding boundary.

The coating protects the glass; index contrast guides the light.

## 02 / INCIDENCE AT THE BOUNDARY

$$\theta > \theta_c$$

$$\sin \theta_c = n_2 / n_1$$

$\theta$  is measured from the normal.